individual technique. They then tested to see which components were most crucial to the success of Rainbow, and the results were that distributional Q-learning, multistep Q-learning (covered in chapter 5), and prioritized replay (which will be briefly covered in section 7.7) were the most important to the Rainbow algorithm's performance.

In this chapter you will learn how to implement a distributional deep Q-network (Dist-DQN) that outputs a probability distribution over state-action values for each possible action given a state. We saw some probability concepts in chapter 4, where we employed a deep neural network as a policy function that directly output a probability distribution over actions, but we will review these concepts and go into even more depth here, as these concepts are important to understand in order to implement Dist-DQN. Our discussion of probability and statistics may seem a bit too academic at first, but it will become clear why we need these concepts for a practical implementation.

This chapter is the most conceptually difficult chapter in the whole book as it contains a lot of probability concepts that are difficult to grasp at first. There is also more math here than in any other chapter. Getting through this chapter is a big accomplishment; you will learn or review a lot of fundamental topics in machine learning and reinforcement learning that will give you a greater grasp of these fields.

7.2. Probability and statistics revisited

While the mathematics behind probability theory is consistent and uncontroversial, the interpretation of what it means to say something as trivial as "the probability of a fair coin turning up heads is 0.5" is actually somewhat contentious. The two major camps are called *frequentists* and *Bayesians*.

A frequentist says the probability of a coin turning up heads is whatever proportion of heads are observed if one could flip the coin an infinite number of times. A short sequence of coin flips might yield a proportion of